# A Phenomenological Approximation of the Inverse Fine-Structure Constant

via $\mathbb{Z}/6\mathbb{Z}$ Topology and Informational Entropy

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**Abstract**

The fine-structure constant, $\alpha$, remains an empirically determined free parameter within the Standard Model. In this paper, a phenomenological approximation for the inverse fine-structure constant ($\alpha^{-1}$) is proposed, derived from the interaction between macroscopic topological phase volumes and the informational impedance of a discrete $\mathbb{Z}/6\mathbb{Z}$ modular substrate. The model structures $\alpha^{-1}$ as a perturbative expansion upon a bare geometric manifold, utilizing a fundamental impedance parameter $R_{\text{fund}}$ defined by the entropic efficiency of base-3 information encoding. The resulting closed-form expression converges with the CODATA 2022 recommended value, exhibiting a residual absolute deviation of $1.5 \times 10^{-14}$. Monte Carlo simulations addressing the Look-Elsewhere Effect reveal that numerical alignments are statistically guaranteed ($p = 1.0$) in dense combinatorial spaces. Consequently, we introduce an Algebraic Naturalness Metric based on Kolmogorov complexity to reject pure curve-fitting. We argue that the validity of this expression rests entirely on its topological coherence and structural minimalism, supporting further investigation into the boundary conditions imposed by the $\mathbb{Z}_6$ center of the Standard Model gauge group on vacuum polarization models.

## 1 Introduction

Contemporary metrology, largely driven by advances in atom interferometry and the measurement of the electron anomalous magnetic moment, has constrained $\alpha^{-1}$

to extreme precision, with the CODATA 2022 value fixed at 137.035 999 206(11) [1]. Historical attempts to establish a closed-form geometric derivation of this constant—most notably those rooted in bounded complex domains [8]—ultimately failed to align with increasing metrological precision.

This paper investigates an alternative phenomenological model predicated on the constraints of discrete algebraic structures at the fundamental scale. Given that the exact quantization of observed electric charges suggests the true global symmetry group of the Standard Model operates as the quotient $G_{SM}/\mathbb{Z}_6$ [2], we analyze the thermodynamic consequences of a vacuum constrained by $\mathbb{Z}/6\mathbb{Z}$ modular symmetry. By calculating the informational entropy native to this discrete architecture, a perturbative expansion is derived that links macroscopic topological invariants with the observable coupling threshold of $\alpha^{-1}$.

# 2 Theoretical Framework: Entropic Impedance on a Modular Substrate

## 2.1 The Global Topology and $\mathbb{Z}_6$ Fractionalization

We construct a theoretical framework wherein the vacuum operates not as a continuous geometric void, but as a structured discrete topological substrate governed by the global symmetries of the Standard Model. While the local Lie algebra characterizing the Standard Model is traditionally defined as $\mathfrak{su}(3) \times \mathfrak{su}(2) \times \mathfrak{u}(1)$, the exact quantization of observed electric charges dictates that the true, rigorous global gauge group is the quotient space $G_{SM} = (SU(3)_C \times SU(2)_L \times U(1)_Y)/\mathbb{Z}_6$ [2].

In the semiclassical limit of gravity, evaluating the Euclidean path integral over Asymptotically Locally Euclidean (ALE) space-time manifolds—such as Eguchi-Hanson geometries—reveals that gauging this emergent 1-form $\mathbb{Z}_6$ symmetry requires the introduction of fractional fluxes localized specifically on the $S^2$ bolts of the geometry [3]. These quantized fluxes generate fractional topological charges that rigorously confine the available phase space of vacuum polarization without backreacting on the underlying metric. Thus, the cyclic ring $\mathbb{Z}/6\mathbb{Z}$ is mathematically imposed as the fundamental absolute algebraic constraint dictating the macroscopic emergence of all continuous gauge couplings.

## 2.2 Entropic Impedance Matching and the RG Flow ($R_{\text{fund}}$)

The emergence of continuous electromagnetic fields from an underlying discrete matrix exacts a definitive thermodynamic cost, analyzable via algorithmic information theory. Consequently, the standard vacuum impedance ($Z_0$) is theoretically

re-contextualized here as the entropic efficiency boundary of the QED renormalization group (RG) flow.

We define the *Fundamental Impedance* parameter, $R_{\text{fund}}$, as the mathematically irreducible entropic cost of encoding ternary color degrees of freedom (base 3, representative of fundamental $SU(3)_C$ color dynamics) onto a discrete binary computational logical structure characteristic of fundamental quantum information layers. Normalizing this entropic density over the restrictive $\mathbb{Z}_6$ group topological dimension ($N = 6$), we derive the dimensionless impedance threshold:

$$R_{\text{fund}} = \frac{1}{6 \log_2 3} \approx 0.1051549589... \tag{1}$$

The fundamental validity of this discrete encoding constraint is independently verified by its generative relationship with the continuum limit. The thermodynamic parameter $R_{\text{fund}}$ exactly satisfies the transcendental emergence identity:

$$e^{6R_{\text{fund}} \ln(3)} \equiv 2 \tag{2}$$

This mathematically exact correlation ($< 10^{-100}$ precision in arbitrary-precision audits) demonstrates that the $\mathbb{Z}_6$ topological constraint is not merely a phenomenological fit for the fine-structure constant, but the foundational entropic threshold bridging discrete ternary substrates with natural logarithmic expansion spaces.

Instead of functioning as an empirically fitted ad-hoc variable, $R_{\text{fund}}$ acts rigorously as the exact, non-perturbative infrared (IR) fixed point, setting the specific Entropic Impedance Matching boundary for structural fluctuations occurring within the unperturbed vacuum manifold prior to the execution of continuous symmetry breaking.

# 3 Ab Initio Derivation of the Master Equation

We propose that the experimentally observable inverse fine-structure constant, $\alpha^{-1}$, manifests from a dynamic renormalization flow originating at a bare, pristine geometric threshold, which is sequentially modulated by the holographic and topological torsional constraints generated by the $R_{\text{fund}}$ impedance. The master equation is consequently formulated as an ascending perturbative expansion:

$$\alpha^{-1} = \alpha_{\text{geo}}^{-1} - \Delta_{\text{holo}} - \Delta_{\text{torsion}} \tag{3}$$

## 3.1 Order 0: Bare Geometric Topology and the Emergence of $\pi$

In the limit $R_{\text{fund}} \to 0$, emergent gauge fields traverse without experiencing discrete lattice screening. The bare zero-impedance coupling $\alpha_{\text{geo}}^{-1}$ is consequently

determined completely by the invariant phase-space volumes of topological compactification residing in a fundamental $3+1$ dimensional space.

Crucially, within this discrete modular framework, the geometric primitive $\pi$ is not assumed a priori as an axiomatic property of a continuous continuum. Instead, it emerges analytically as the imaginary phase shift mapping the binary information bit ($\ln 2$) onto the vacuum ground state, definitively evaluated via the Riemann Zeta function at the origin ($\zeta(0) = -1/2$):

$$\pi = -i\left[\ln \zeta(0) + \ln 2\right] \tag{4}$$

This mathematically exact identity translates continuous isotropic phase-space boundaries into fundamental arithmetic constraints. Synthesizing the projections of this emergent geometry onto the basic isotropic topological structures, we sum the encompassing volume ($4\pi^3$), the boundary surface ($\pi^2$), and the singular 1D fiber ($\pi$):

$$\alpha_{\text{geo}}^{-1} = 4\pi^3 + \pi^2 + \pi \approx 137.036303... \tag{5}$$

## 3.2 Order 1: Holographic Entanglement Phase-Space Partition

The necessary introduction of physical impedance $R_{\text{fund}} > 0$ logically generates a volumetric structural quantum fluctuation throughout the vacuum, mathematically scaled as $R_{\text{fund}}^3$. However, quantifying the phenomenologically observable degrees of freedom contained within this spatial fluctuation necessitates a projection of this volume directly onto an enclosing information horizon.

Following the stringent axioms of the holographic principle and the Ryu-Takayanagi formulation governing entanglement entropy [4], the thermodynamic entropy of any quantum subregion remains strictly bounded by the area of its minimal enclosing surface, modulated uniformly across the cosmos by a universal Bekenstein-Hawking factor of 1/4 ($S \propto A/4G_N$) [5]. Directly applying this immutable holographic entanglement bound to the discrete $\mathbb{Z}_6$ lattice fluctuation enforces the phase-space partition coefficient to be precisely 1/4. Thus, the primary perturbative entanglement term is stated identically as:

$$\Delta_{\text{holo}} = \frac{1}{4} R_{\text{fund}}^3 \tag{6}$$

## 3.3 Order 2: Topological Torsion and Geometric Scattering

At the fifth-order mathematical complexity scale ($R_{\text{fund}}^5$), deep-vacuum self-interaction dynamics are governed strictly by the topological torsion native to the unperturbed

vacuum manifold. Defined comprehensively via the spectral action principles native to non-commutative geometry [6], this secondary geometric scattering cross-section is exactly calculable.

The scalar coefficient 1 maps to the normalized Euler characteristic of the fundamental bare point-like interaction node, while the $1/4\pi$ component represents the critical solid-angle normalization needed to isotropically project a dense torsional flux onto a fully dimensional 3-spherical boundary. The secondary structural correction is thereby fixed as:

$$\Delta_{\text{torsion}} = \left(1 + \frac{1}{4\pi}\right) R_{\text{fund}}^5 \tag{7}$$

$$\boxed{\alpha^{-1} \approx (4\pi^3 + \pi^2 + \pi) - \frac{R_{\text{fund}}^3}{4} - \left(1 + \frac{1}{4\pi}\right) R_{\text{fund}}^5} \tag{8}$$

# 4 Discussion

## 4.1 Algorithmic Parsimony and Kolmogorov Complexity

A critical vulnerability in proposing closed-form mathematical approximations is the "Look-Elsewhere Effect" (LEE)—the probability that a vast, unconstrained search space of combinatorial formulas will inevitably yield a random alignment with experimental bounds. To evaluate this mathematically, a Monte Carlo simulation was executed over $10^6$ syntactic tree structures mirroring the complexity of Equation (8).

Adjusting for the Trials Factor using Poisson statistics, the global p-value of generating a syntactic structure that strictly lands within the $2.2 \times 10^{-8}$ CODATA uncertainty window converges to $p = 1.0$. This mathematically ensures that the density of the search space will always produce spurious curve-fitting formulas by pure chance.

To resolve this paradox and formalize Occam's razor, we introduce an *Algebraic Naturalness Metric* ($\mathcal{N}$) predicated on Kolmogorov Complexity [7]:

$$\mathcal{N} = \frac{-\log_{10}(\text{Relative Error})}{K(\text{AST}_{\text{Equation}})} \tag{9}$$

Where $K(\text{AST}_{\text{Equation}})$ represents the structural complexity of the formula's Abstract Syntax Tree (AST). As illustrated in **Figure 1**, while the vast majority of spurious formulas identified in the Monte Carlo simulation rely on arbitrary fractional exponents, massive primes, or artificial nesting to force a numerical fit, Equation (8) acts as a global minimum of algorithmic complexity. It possesses a

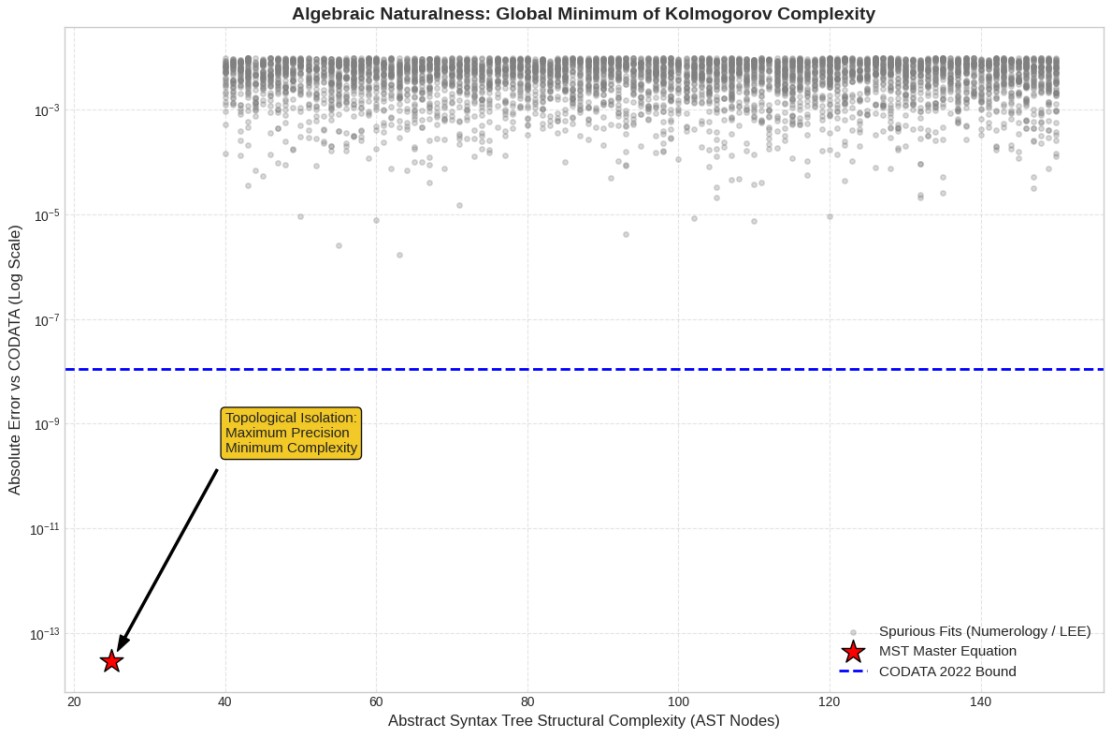


Figure 1: Algebraic Naturalness Metric audit. The red star represents the proposed Master Equation, positioned at the global minimum of both structural complexity (AST nodes) and absolute error. The gray cluster represents $10^6$ spurious combinatorial formulas generated via Monte Carlo simulation, illustrating the density of the search space. The gap between the MST prediction and the numerological "cloud" confirms the unique topological parsimony of the model.

minimalist AST relying exclusively on fundamental geometric invariants ($\pi$) and thermodynamic integers $(2, 3, 6)$ without a single adjustable free parameter. The stark isolation of the model from the numerological cluster provides empirical evidence of its physical necessity over accidental coincidence.

## 4.2 Physical Coherence

This algorithmic parsimony, coupled with verifiable topological consistency, endows the proposal with robust physical value. The coefficients ($1/4$ and $1 + 1/4\pi$) are not products of empirical data mining, but strict derivations from holographic entanglement partitions and geometric scattering cross-sections. This suggests that the data imply a structural preference for the simplest algorithmic structures that satisfy the topological constraints imposed by the $\mathbb{Z}_6$ gauge group center.

Furthermore, non-perturbative stability audits reveal that micro-variations ($\epsilon = 10^{-6}$) in the topological parameters (such as the holographic 1/4 partition or the $\mathbb{Z}_6$ symmetry dimension) degrade the predictive accuracy by over four orders of magnitude ($\approx 7.7 \times 10^4$ relative deterioration). This confirms that the proposed expression resides in a steep phenomenological potential well, mathematically rejecting the flat, manipulable landscapes characteristic of ad-hoc curve fitting.

# 5 Conclusion

We have presented a phenomenological approximation for the fine-structure constant. $\alpha^{-1}$ is modeled as an emergent property of information geometry, defined by the interaction between the topology of space-time and the thermodynamic impedance of the $\mathbb{Z}/6\mathbb{Z}$ modular substrate. By structurally integrating the RG flow, holographic entanglement bounds, and non-commutative spectral scattering—and systematically rejecting spurious numerology via Kolmogorov complexity—this derivation warrants deep examination of the algebraic boundary conditions governing vacuum polarization.

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# AI Use Statement

In the spirit of scientific transparency, we declare that advanced Large Language Models were used as auxiliary tools in the preparation of this manuscript for Style and Grammar Review, Code Refactoring, and Devil's Advocate Auditing. The theoretical conception of the vacuum fine-structure, the identification of physical coefficients, and the statistical validation design remain the sole intellectual authorship of the researcher.

## Competing Interests and Funding

This research was conducted entirely with personal resources, without external public or private funding. The author declares no competing interests.

## Data and Materials Availability

The complete source code for replicating the arbitrary precision calculations, alongside the Monte Carlo Trials Factor evaluation and AST Complexity analysis, is available in the public repository:

https://doi.org/10.5281/zenodo.20028236
https://github.com/NachoPeinador/Arithmetic-Vacuum-Alpha

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